2.8 Task 0 – Check your environments

Run the cell below in **Google Colab and in one desktop tool** of your choice. Paste a screenshot of both outputs into the work area.

```
[5]
✓ 0s
# !pip install ISLP
# [ADDED] worked example – run this cell
# Environment check – run in each environment and include the output in your report
import sys, platform, importlib
print("Python version :", sys.version.split()[0])
print("Executable     :", sys.executable)
print("Operating sys. :", platform.system(), platform.release())
for pkg in ["numpy", "pandas", "matplotlib", "sklearn", "statsmodels", "ISLP"]:
    try:
        m = importlib.import_module(pkg)
        print(f"{pkg:<12} {getattr(m, '__version__', 'installed')}")
    except ImportError:
        print(f"{pkg:<12} NOT INSTALLED – run the Setup cell / pip install ISLP")
```

```
Python version : 3.13.16
Executable     : /usr/bin/python3
Operating sys. : Linux 6.6.122+
numpy        2.1.3
pandas       2.2.3
matplotlib   3.10.0
sklearn      1.6.1
statsmodels  0.15.0
ISLP         0.4.1
```

```
# [ADDED] worked example – run this cell
# Environment check – run in each environment and include the output in your report
import sys, platform, importlib
print("Python version :", sys.version.split()[0])
print("Executable     :", sys.executable)
print("Operating sys. :", platform.system(), platform.release())
for pkg in ["numpy", "pandas", "matplotlib", "sklearn", "statsmodels", "ISLP"]:
    try:
        m = importlib.import_module(pkg)
        print(f"{pkg:<12} {getattr(m, '__version__', 'installed')}")
    except ImportError:
        print(f"{pkg:<12} NOT INSTALLED – run the Setup cell / pip install ISLP")
```

[131] ✓ 0.0s Python

```
Python version : 3.14.3
Executable     : c:\Program Files\Python314\python.exe
Operating sys. : Windows 11
numpy        2.4.4
pandas       3.0.2
matplotlib   3.10.8
sklearn      1.8.0
statsmodels  0.15.0
ISLP         0.4.1
```

Exercise 1

(a)–(b): Vector v was generated with even numbers up to 40 and reshaped into a 4×5 matrix M. The matrix has shape $(4, 5)$, 2 dimensions, integer type, and its transpose $M.T$ swaps rows and columns into a 5×4 layout

(c)–(d): Slicing accurately extracts the required rows and columns using 0-based indexing. Aggregations using axis=1 compute horizontal row means, while axis=0 computes vertical column standard deviations

(e): Slicing creates a view sharing memory with M, so modifying B alters M. Conversely, .copy() allocates an independent memory block, protecting the original matrix

Exercise 2

(a)–(b): Data was simulated with $n = 100$ following $x \sim N(0, 1)$ and noise $\varepsilon \sim N(0, 0.5^2)$. The correlation coefficient between x and y is very high (close to 0.98), reflecting a strong linear relationship.

(c)–(d): The scatter plot clearly shows the positive linear trend aligning closely with the red population line. Setting a random seed (rng) ensures full reproducibility; removing the seed would yield different randomized datasets on every run.

Exercise 3

(a): Exactly 5 rows contained missing values (encoded as ? in horsepower), which were successfully removed using .dropna(), leaving 392 observations.

(b)–(c): Variables like mpg, weight, and horsepower are quantitative, whereas name and origin are qualitative. The summary table provides precise ranges, means, and standard deviations for all numerical features after cleaning.

Exercise 4

Removing a consecutive block of observations (from the 10th to 85th row) slightly shifts the mean and standard deviation values of quantitative variables (such as `horsepower` and `weight`).
Variables with wider original spreads experience minor statistical shifts because removing a specific slice of ordered rows alters the empirical distribution of the dataset.

Exercise 5

Predictive Power: Variables like horsepower, weight, and displacement show strong negative correlations with mpg (heavier/more powerful cars have lower fuel efficiency).

Non-linearity: The scatter plots reveal curved (non-linear) patterns between mpg and predictors like horsepower, suggesting that transformations or polynomial terms may be required for linear regression.

Group Differences: Box plots indicate clear variations in mpg across different regions (origin) and engine types (cylinders).

Exercise 6

(a)–(c): The dataset contains 777 US higher education institutions, categorized into private and public. Creating the Elite feature identifies a subset of competitive schools where over 50% of incoming students rank in the top 10% of their high school class.

(d)–(e): Box plots show that private and elite colleges typically charge significantly higher out-of-state tuition (Outstate). Histograms demonstrate right-skewed distributions for application and enrollment metrics across universities.

Exercise 7
Using a for loop combined with f-string formatting ({col:<12}, {mean_val:.2f}, {share:.1%}) provides a clean, automated way to inspect summary metrics across multiple features.

The above-mean share indicates the percentage of observations lying above the arithmetic mean, helping identify skewed distributions (e.g., values around 35%–40% indicate right-skewness).